BODESA

FORECAST DEMAND

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Introduction and Context

Project Objective

Predict the weekly sales for the year **2026** for each product, based on past sales data from **January 2023 to May 2025**.

Detailed Modeling Strategy

1. Data Preparation
2. Models Tested
3. Evaluation

Data Preparation

- Removed rows with negative values
- Filtered products suitable for modeling
- Visualized sales trends to detect patterns and anomalies

Variables descriptions

	0
Date	object
Department	int64
Category	int64
Brand	int64
Sales (Units)	int64
Sales (Amount)	float64
Inventory (Units)	int64

dtype: object

	0
Date	0
Department	0
Category	0
Brand	0
Sales (Units)	0
Sales (Amount)	0
Inventory (Units)	0

dtype: int64

```
[8] data['Department'].nunique()
56
[9] data['Brand'].nunique()
2323
[10] data['Category'].nunique()
2763
[98] data['Date'].nunique()
126
```

Statistics descriptives

	Department	Category	Brand	Sales (Units)	Sales (Amount)	Inventory (Units)
count	1242788.00	1242788.00	1242788.00	1242788.00	1242788.00	1242788.00
mean	23.96	240024.62	3949.67	4.15	3697.91	79.32
std	17.12	171152.66	2698.67	17.76	32858.05	263.41
min	1.00	10001.00	4.00	-43.00	-251721.64	-4958.00
25%	11.00	110108.00	1145.00	0.00	0.00	5.00
50%	21.00	210103.00	3838.00	1.00	178.44	18.00
75%	36.00	361818.00	6542.00	3.00	1612.70	65.00
max	66.00	660504.00	8219.00	3262.00	13197240.68	17871.00

```
[13] int((data['Sales (Units)']<0).sum())
```

4105

```
[14] int((data['Sales (Amount)']<0).sum())
```

5747

```
[15] int((data['Inventory (Units)']<0).sum())
```

23410

			0
Department	**Category**	**Brand**	
1	**10001**	**3525**	126
9	**90220**	**836**	126
		2844	126
	90222	**944**	126
		2129	126
...	**...**	**...**	...
6	**60105**	**8186**	1
31	**310204**	**5993**	1
41	**411007**	**878**	1
9	**90536**	**7133**	1
12	**120206**	**7888**	1

17680 rows × 1 columns

dtype: int64

Assumptions

- There are no product returns from customers.
- The company does not issue any refunds after a sale.
- The sales system does not allow the sale of a product when its stock is zero or insufficient.

Removing rows containing negative values

```python
# Remove rows with negative values in the target columns.
data_t = data[
    (data['Sales (Units)'] >= 0) &
    (data['Sales (Amount)'] >= 0) &
    (data['Inventory (Units)'] >= 0)
]
```

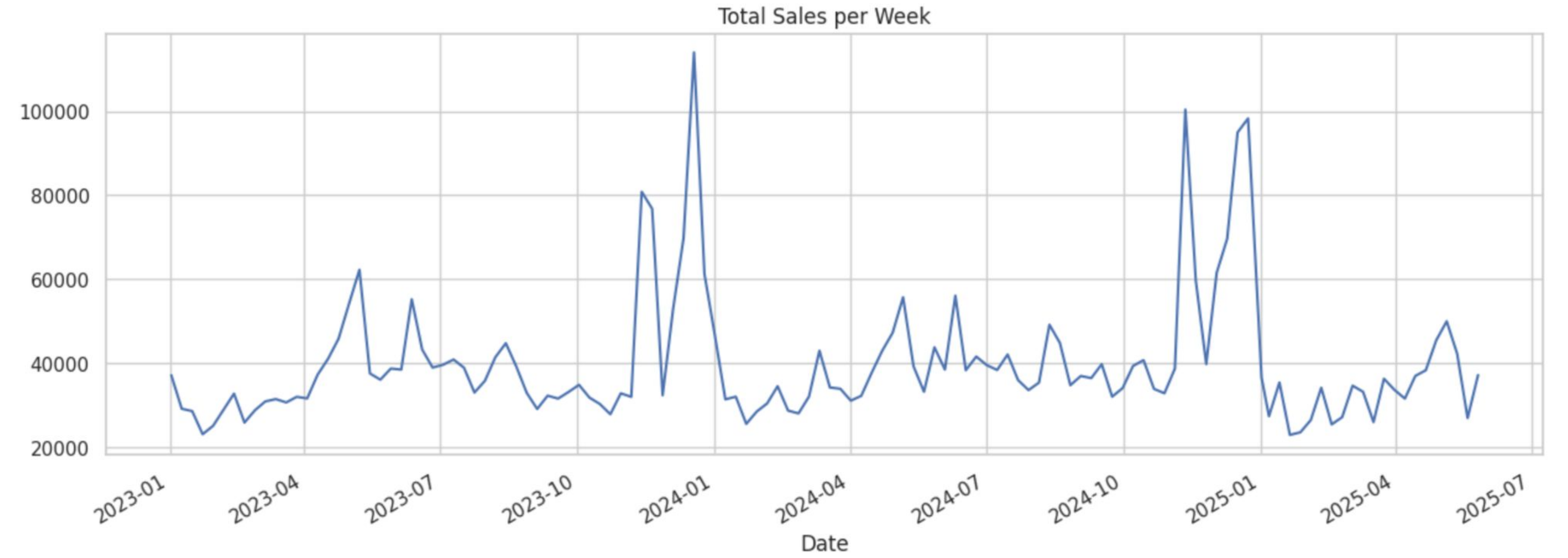
	0
Product_Id	
P_10I100701I6347	126
P_9I90529I6758	126
P_9I90529I6098	126
P_9I90528I6760	126
P_9I90512I6760	126
...	...
P_21I210103I7197	1
P_40I400105I5337	1
P_41I410101I690	1
P_40I400010I1572	1
P_12I120408I606	1

17457 rows × 1 columns

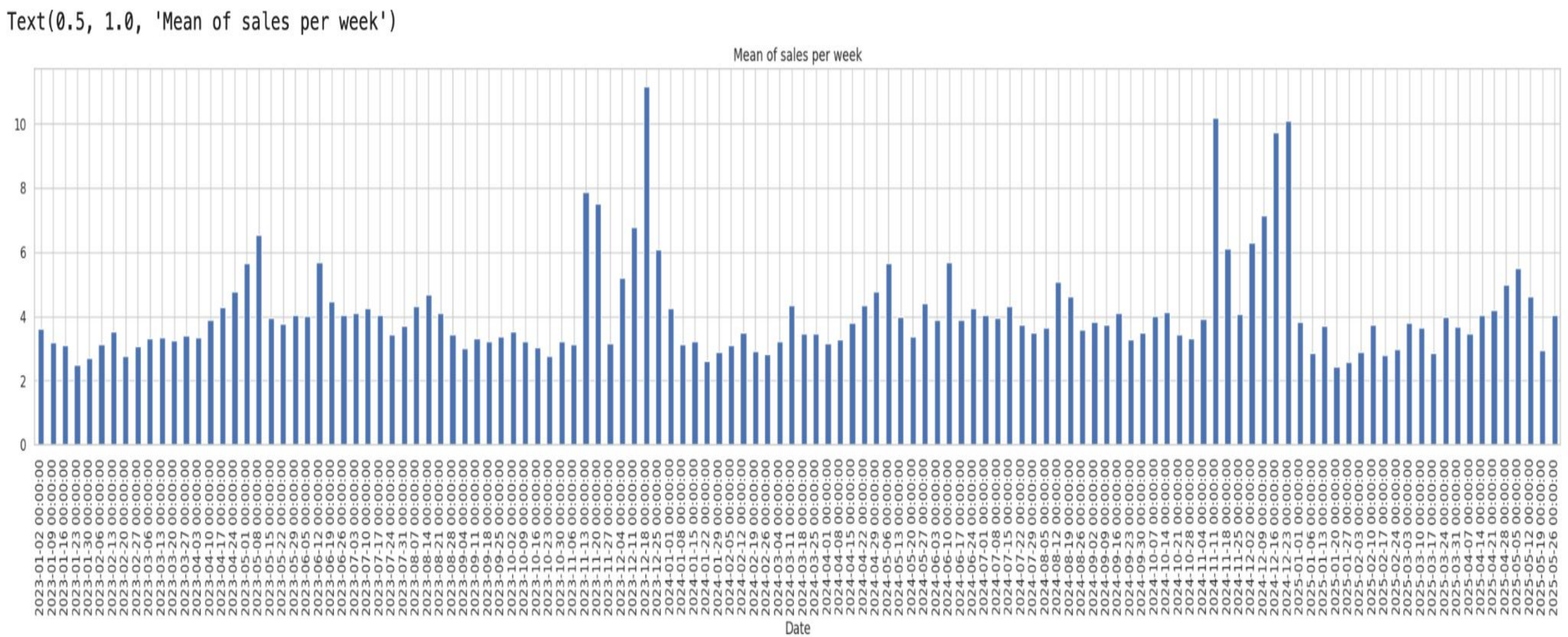
dtype: int64

Graphical Visualization of the Data

```
Text(0.5, 1.0, 'Total Sales per Week')
```



Graphical Visualization of the Data



Filtering of modelizable products

```python
def is_difficult_series(group):
    sales = group['Sales (Units)']
    return (
        len(sales) < 104 or
        sales.std() == 0   or
        sales.autocorr(lag=1) < 0.3 or
        (sales == 0).sum() / len(sales) > 0.6
    )

produits_difficiles = (
    data_t.groupby(['Department', 'Category', 'Brand'])
    .filter(is_difficult_series)
)
```

```python
data1 = (
    data_t.groupby(['Department', 'Category', 'Brand'])
    .filter(lambda g: not is_difficult_series(g))
)
```

	0
Product_Id	
P_9I90222I5875	126
P_11I110102I1228	126
P_9I90681I3792	126
P_9I90222I4490	126
P_11I110102I5676	126
...	...
P_47I470402I1016	104
P_45I450401I7797	104
P_46I460001I6251	104
P_11I110114I4340	104
P_45I450201I7206	104

4070 rows × 1 columns

dtype: int64

Models Tested

ARIMA – Captures general trends based solely on historical data

SARIMA – ARIMA extended with seasonality handling

SARIMAX – SARIMA + external variables (e.g., promotions, holidays)

Prophet – Flexible model from Facebook, detects trends and seasonality automatically

Prophet + XGBoost – Combines Prophet for seasonality with XGBoost for extra predictive power using external features

Test

Thank you for your attention.

Gracias por su atención.

Merci pour votre attention.